



Universität Hamburg
DER FORSCHUNG | DER LEHRE | DER BILDUNG

Master Thesis

The impact of tropospheric versus stratospheric CO₂ forcing on the Hadley cell

Ole Merkes

Atmospheric Science (M.Sc.)
Matr.-Nr.: 7830665

First Supervisor: Prof. Dr. Sarah Kang
Second Supervisor: Dr. Moritz Günther

The impact of tropospheric versus stratospheric CO₂ forcing on the Hadley cell

Master Thesis

by

Ole Merkes

Student Name	Student Number
--------------	----------------

Ole Merkes	7830665
------------	---------

Supervisors: Prof. Dr. Sarah Kang
Dr. Moritz Günther



Universität Hamburg
DER FORSCHUNG | DER LEHRE | DER BILDUNG

Abstract

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetur adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

Contents

Summary	i
Nomenclature	iii
1 Introduction	1
2 Theory	3
2.1 The atmosphere under greenhouse gas forcing	3
2.2 Hadley circulation strength under greenhouse gas forcing	3
2.3 The angular momentum budget	4
2.3.1 Axisymmetric theory	4
2.3.2 Eddy driven mean circulation	6
2.4 The energy budget of the atmosphere	7
2.5 Model setup	11
2.5.1 ECHAM5	11
2.5.2 konrad	12
3 Methods	13
3.1 ECHAM5	13
3.2 konrad	13
3.3 Decomposition in forcing factors	13
4 Results	14
4.1 The atmospheric response	14
4.2 Predictions from axisymmetric scalings	15
4.3 Eddy driven Hadley cell predictions	17
4.4 The energetic framework	18
4.4.1 GMS and energy transport of the Hadley cell	18
4.4.2 Eddy energy transport	19
4.4.3 Forcing and feedback contributions	20
4.4.4 The forcing	20
5 Discussion	22
6 Conclusion	23
References	25

Nomenclature

Abbreviations

Abbreviation	Definition
USA	United States of America
EU	European Union

Variable Names

Symbol	Definition	Unit
α	greek alpha	[N]
β	greek beta	[J]

1

Introduction

The Hadley cell (HC) is the dominant meridional large-scale circulation in the tropics. Its ascending branch at the equator is collocated with inter-tropical convergence zone (ITCZ), which is characterized by large amounts of precipitation. On the other hand, the dry, semi-arid and arid regions of subsidence in the subtropics are associated with the descending branch of the HC. In the annual mean and in the equinoxes there are two similar overturning cells in the tropics - one in each hemisphere. It extends from 0 to 30 °N in the Northern hemisphere and equivalently 0 to 30 °S in the Southern hemisphere. Throughout the annual cycle there are strong variations in the HC's extent and strength: As the maximum of insolation moves poleward, so does the ITCZ and the HCs, resulting in a strong cross-equatorial winter (hemisphere without the maximum) and a weaker summer HC (hemisphere with the maximum). This seasonal migration is related to the global monsoon, which has been identified as an angular momentum conserving HC regime (Bordoni & Schneider, 2008).

The HC is responsible for the poleward transport of energy and the equatorward transport of moisture, thereby reducing the meridional temperature gradients imposed by solar insolation. The HC has strong implications for human and animal life as well as agriculture and vegetation on this planet. Its extent crucially modulates global precipitation patterns by setting the regions of ascend and descend. The strength of the HC determines what amount of energy is reaching the off-equatorial region and what amount of moisture is reaching the equator, setting the potential precipitation amounts. This is particularly evident in the Sahel region, which is located approximately 10 °N of the equator on the African continent. This semi-arid region receives most of its precipitation in the boreal summer when the HC and the ITCZ migrate northward. Thus, the rainfall in the Sahel is dynamically linked to the HC position. Changes in the HC also affect the livelihood of the local population and economies that remain heavily dependent on agricultural production.

This has also been shown for paleoclimate. Paleoclimate simulations of the HC display large changes in HC strength and extend during the last glacial maximum and the mid holocene due to changes in orbital forcing (D'Agostino et al., 2017). This is also supported by evidence from tropical paleoclimate proxy records of precipitation changes (mcgeeChangesITCZLocation2014). This past evidence of strong HC changes combined with the large societal impact of precipitation patterns give a strong reason to investigate the projected HC changes of human driven greenhouse gas emissions in the current era.

The CO₂ in the atmosphere is very uniformly distributed. The forcing that it imposes is not. The forcing is critically set by the difference between surface and stratospheric temperature,

which varies strongest in the meridional direction. Given that the HC is flattening the meridional energy gradient, the response of the HC strength depends on meridional forcing structure. This has been shown for quadrupling of CO₂ only in certain latitude belts in coupled simulations (Kim et al., 2022). A forcing in mid latitudes weakens the Hadley circulation whereas a forcing at the equator leads to it strengthening. We investigate how the *vertical* structure of CO₂ forcing, rather than its horizontal structure, alters atmospheric circulation. We achieve this by running simulations with a General Circulation Model (GCM) while quadrupling CO₂ only in the troposphere and only in the stratosphere using a pre-industrial control tropopause height to separate the two cases. To our knowledge, these experiments have not yet been described by the literature.

Climate simulations from the Coupled Model Intercomparison Project (CMIP), have established a HC weakening and widening under climate change (Lu et al., 2007; Xia et al., 2020). Explaining this, previous studies have emphasized the importance of drivers related to the troposphere (Chemke & Polvani, 2021; Fan & Dommenges, 2024). However, historical simulations show that stratospheric forcing also influences the general circulation. In particular stratospheric ozone depletion has led to a cooling of the Southern hemisphere lower-stratospheric polar cap. This has been identified as the main driver for the widening of the Hadley circulation over the second half of the twentieth century (Polvani et al., 2011). The goal of our study is to disentangle the contributions of tropospheric and stratospheric CO₂ to the established HC weakening with uniform CO₂ distribution. We find that the HC response to tropospheric CO₂ is a strengthening and widening in both hemispheres. This opposes our expectations evoked from the literature, which identifies tropospheric warming parameters as the main drivers for HC weakening under uniform CO₂ concentrations. Our results suggest that the weakening of the HC originates from stratospheric cooling caused by stratospheric CO₂. To investigate and explain the different responses of the HC strength we employ angular momentum and energy budget frameworks. We use the axisymmetric angular momentum framework to attribute HC changes to changes in tropopause height, meridional temperature, gradient, and vertical stability. Since this theory ignores the contribution of eddies to HC strength we also investigate changes in eddy divergence to explain the observed changes in the mean circulation. The energetic framework is a diagnostically unifying approach. It uses changes in top-of-atmosphere (TOA) radiative fluxes and moist static energy to distinguish different contributing factors to HC changes.

We contribute to the advancement of the field by revising the applicability of different HC frameworks and simplified models. Splitting up CO₂ forcing in two vertical layers puts existing theories to the test and gives the opportunity to rethink accepted truths in climate science. Overall, this is a contribution to foundational research that advances our physical understanding of the drivers of general circulation changes.

TBD Overview over the study Is this really necessary? EWE winds

2

Theory

2.1. The atmosphere under greenhouse gas forcing

climate sensitivity (Jeevanjee, 2023) radiative equilibrium forcing and feedback tropospheric warming Why is the troposphere warming under greenhouse gas forcing? What is the warming structure? How can we explain it? Enhanced upper tropospheric warming in the tropics Arctic amplification

stratospheric cooling Why is the stratosphere cooling under greenhouse gas forcing?

2.2. Hadley circulation strength under greenhouse gas forcing

The HC can be characterized by its strength and extent. The HC can be described by the mass stream function (MSF)

$$\psi_{\max} = \frac{2\pi a \cos \varphi}{g} \int_{p_s}^0 \bar{v} dp'. \quad (2.1)$$

The MSF is always parallel to the flow in the vertical p - φ plane. A positive value of $\psi_{\max}$ indicates a clockwise circulation and a negative value a counter-clockwise circulation. In the annual mean, the Northern hemisphere shows a clockwise circulation and a counter-clockwise circulation in the Southern hemisphere. Since the Hadley cell vertically integrates the mass flux, its values symbolize the net magnitude of mass transport above and below. And the HC maximum the integral covers exactly the upper or lower branch of the circulation. In the literature one can find several metrics for the HC strength. Some use the MSF at a fixed pressure level or spatially average, while others use the minimum of the vertical pressure tendencies ω . HC trends have show a large sensitivity to the metric used (Pikovnik et al., 2022). We find it most plausible to calculate the strength of the HC from the maximum of the $\psi_{\max}$ and the extent from the latitude at which $\psi_{\max}$ becomes zero, averaged between 300 and 700 hPa.

Direct effect of CO₂ forcing on the HC (Bony et al., 2013) Quantifying the change of the HC extent and strength under climate change (Lu et al., 2007) (Vallis et al., 2015) What is the role of different feedback mechanisms in shaping the HC response to climate change (Feldl & Bordoni, 2016) Evaluation of different drivers of HC changes (Chemke & Polyani, 2021)

Which drivers have been identified for the weakening of the HC

2.3. The angular momentum budget

Considering the angular momentum budget of the Earth, one addresses the fundamental question of why there is a meridional circulation in the atmosphere in the first place. An answer is provided by the axisymmetric theory of Held and Hou (1980). It also provides a scaling for the HC strength and extent linking it directly to thermodynamical parameters. Because this theory ignores eddies, we are further investigating their effect on the HC in the following subsection. Since there is no unifying theory incorporating angular momentum conservation and angular momentum transport by baroclinic eddies, we introduce the energetic budget framework of the HC to explain why it changes. This is a more diagnostic framework which does not explain the existence of the HC from first-hand principles but rather takes the necessity transport energy poleward as granted. This necessity can be inferred from observational data of the TOA radiative imbalance showing an excess of incoming radiative fluxes relative to the outgoing radiation at the equator and vice versa at higher latitudes. This transport by the the atmosphere can be split up into a mean component, i.e., the HC in the tropics, and and eddy component.

2.3.1. Axisymmetric theory

The axisymmetric theory considers a flow in the φ - z plane with no meridional circulation ($v = w = 0$). This prevents the existence of large scale barotropic or baroclinic eddies. The questions why there is a meridional circulation in the first place is answered by the axisymmetric theory (Held & Hou, 1980). This question is raised because one could imagine a world where Earth is just in radiative equilibrium everywhere. This means that every point on Earth emits just as much radiation as it receives from the sun. Thus, there would not be any meridional temperature or energy transport necessary. Given a radiative-equilibrium temperature profile of the Earth

$$\frac{\theta_E}{\theta_0} = 1 - \frac{1}{3}\Delta_h(3\sin^2\varphi - 1) + \Delta_v\left(\frac{z}{H} - \frac{1}{2}\right) \quad (2.2)$$

(see Figure 2.1a), one can obtain a corresponding zonal wind u_E via the thermal wind relationship. Here, Δ_h is the fractional pole to equator temperature difference, Δ_v the fractional temperature difference between the surface and the top of the fluid, H the depth of the fluid, and θ_0 the global mean value of θ_E . This solution not only produces westerly winds at the equator, which are not observed in the atmosphere, but also violate Hide's theorem (Hide, 1969). Hide's theorem states that a maximum of angular momentum within the flow domain cannot be sustained due to dissipation within the flow. A maximum of angular momentum can only exist at the boundaries. This tells us that this RE temperature profile is not able to persist on Earth and that we require some poleward transport of energy.

The conservation of angular momentum

$$M = (u + \Omega a \cos \varphi)a \cos \varphi \quad (2.3)$$

provides a second constraint for a HC solution with the zonal wind u , a being the radius and Ω the rotational frequency of the Earth. The angular momentum conserving wind can then be derived as

$$u_M = \Omega a \frac{\sin^2 \varphi}{\cos \varphi}. \quad (2.4)$$

This allows us to infer, via the thermal wind relationship, a temperature profile θ_M (see Figure 2.1a), which is now slaved to angular momentum conservation. Assuming that the energy is conserved in the upper branch and that the solution is continuous at the edge of the HC, allows us to determine the undefined parameters of this solution. One also obtains a finite scaling for the HC width

$$\phi_H \sim \left(\frac{gH\Delta_h}{\Omega^2 a^2} \right)^{1/2} \quad (2.5)$$

and for its strength

$$\psi_{\max} \sim \frac{1}{\tau} \frac{g^{3/2} (H\Delta_h)^{5/2}}{\Delta_v a^2 \Omega^3} \quad (2.6)$$

where g is the gravitational acceleration of Earth and τ is the time scale at which θ relaxes to θ_E .

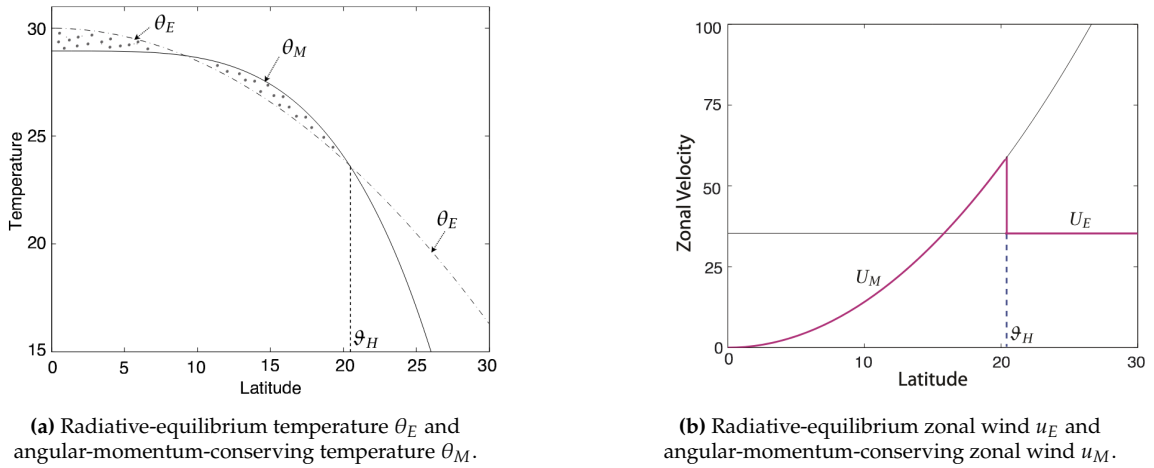


Figure 2.1: Latitudinal profiles of temperature and zonal wind under radiative equilibrium and under angular momentum conservation, adapted from Vallis (2017).

The key contribution of this theory is to explain why there is a HC with finite width in the first place. It also shows that the HC does not extent from the equator to the poles. Subsequent studies have advanced this theory also to temperature profiles which have their maximum off the equator, in order to account for seasonal variations (Lindzen & Hou, 1988). In our axisymmetric model, the HC is weaker in equinox conditions than observed, leading to the suggestion that the HC in the annual mean is not close to equinox conditions but rather that the HC flips between solstitial states. More recent studies have addressed this discrepancy using eddy permitting models (Walker & Schneider, 2005). In these models, the strength of an annually averaged Hadley circulation is comparable to the Hadley circulation driven by an annually averaged heating. This means that the HC in its equinox state resembles very much an annual mean HC. This highlights the importance of eddies in driving the HC strength, which will be covered in the next section.

2.3.2. Eddy driven mean circulation

The importance of eddies in shaping the HC was acknowledged even before the axisymmetric theory was developed (Kuo, 1956). More recent studies have compared axisymmetric models to eddy permitting simulations to determine the importance of eddies in altering HC strength (Singh et al., 2017; Walker & Schneider, 2006). They have found that the eddies increase the HC strength by 50–75%, depending on the model.

From the eddy perspective the HC edge is set by the latitude at which the zonal winds become baroclinically unstable. Baroclinic instability arises due to large vertical shear in a flow. Using a two layer model, one can derive the HC edge as (Held, 2000)

$$\phi_H \sim \left(\frac{NH_e}{\Omega a} \right)^{1/2} \quad (2.7)$$

with N being the Brunt-Väisälä frequency and H_e the depth of the troposphere in the extratropical regions. Correlating the HC edge with tropical tropopause height and extratropical tropopause height in CMIP models has shown that the HC edge is set by the extratropical tropopause height (Lu et al., 2007) instead of the tropical tropopause height. This favors an eddy driven perspective of the HC edge.

To quantify HC strength changes from the eddy perspective, we apply a Reynolds decomposition on the zonal and meridional wind. We obtain the eddies as deviations in time from the zonal and temporal mean flow

$$u = \bar{u} + u', \quad v = \bar{v} + v'. \quad (2.8)$$

These deviations are associated with weather and Rossby waves in the mid-latitudes. Inserting this into the zonal momentum equation and neglecting viscous forces and vertical advection, we obtain

$$f\bar{v}(1 - Ro) = \frac{1}{a \cos^2 \varphi} \frac{\partial(\cos^2 \varphi \overline{u'v'})}{\partial \varphi} + \frac{\partial \overline{u'\omega'}}{\partial p}. \quad (2.9)$$

Here, the Rossby number is

$$Ro = \frac{\bar{\xi}}{f} \quad (2.10)$$

where $\xi = \hat{k} \cdot \nabla \times \vec{u}$ is the relative vorticity. Equation (2.9) has two limits. For $Ro \rightarrow 1$, the left-hand side becomes zero and the eddy term becomes negligible. This means that our Hadley circulation is driven by directly by thermal driving and is close to angular momentum conservation

$$\frac{\partial \bar{M}}{\partial \varphi} = 0 \quad (2.11)$$

as described by the axisymmetric solution in Section 2.3.1. For $Ro \rightarrow 0$, the meridional wind $\bar{v}$ is instead balanced by the eddy momentum flux convergence. This regime can be understood as eddy-driven (see Figure 2.2).

The climatological divergence of eddy momentum flux, which is observed in the tropics, requires a positive mean meridional wind. This holds even if the zones of baroclinic instability lie well poleward of the HC edge because Rossby waves can propagate into the HC, producing a poleward momentum flux. At the HC edge there is $\bar{v} = 0$ and thus no eddy momentum flux. The eddy momentum flux divergence also explains the equatorward meridional wind in the eddy-driven Ferrel cell in mid-latitudes. Over the seasonal cycle, the Rossby number varies between 0.2 and 0.8, approaching either of the two limits. During the equinoxes the Rossby number is small which indicates a strong eddy influence. During the solstices the Rossby number is closer to 1 in the winter HC but still small in the summer HC.

Assuming that the HC is entirely eddy driven ($Ro = 0$), the HC strength can be predicted from the eddy momentum flux by inserting Equation (2.9) into the Equation (2.1). Following Walker and Schneider (2006), this gives the integrated eddy momentum flux divergence

$$\psi_{\max} = \frac{2\pi a \cos \varphi}{f g} \int_{p_{\max}}^0 \left(\frac{1}{a \cos^2 \varphi} \frac{\partial(\cos^2 \varphi \overline{u'v'})}{\partial \varphi} + \frac{\partial \overline{u'\omega'}}{\partial p} \right) dp' \equiv \mathcal{T} \quad (2.12)$$

with $p_{\max}$ being the pressure level at which the MSF becomes maximum and the Coriolis parameter f . The quantity $\mathcal{T}$ is a predictor for the HC strength given a certain eddy momentum flux divergence. This predictor has been shown to be skillful in predicting the HC strength in eddy permitting models for various climates (Walker & Schneider, 2006).

DRY!!!

2.4. The energy budget of the atmosphere

The energy budget analysis provides diagnostic perspective on the HC. It cannot explain why the Earth is not in radiative (convective) equilibrium but assumes the necessity for poleward transport as given. Using TOA radiative fluxes (see Figure 2.3), one can infer that the short-wave radiative flux at the equator is much larger than the emitted outgoing long wave radiation. The opposite is true for the poles. This implies an poleward energy transport by the atmosphere and the ocean.

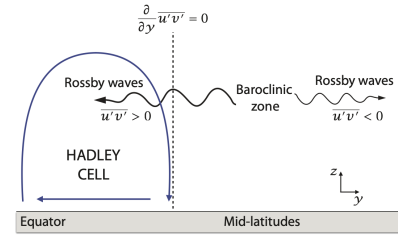


Figure 2.2: (Vallis, 2017)

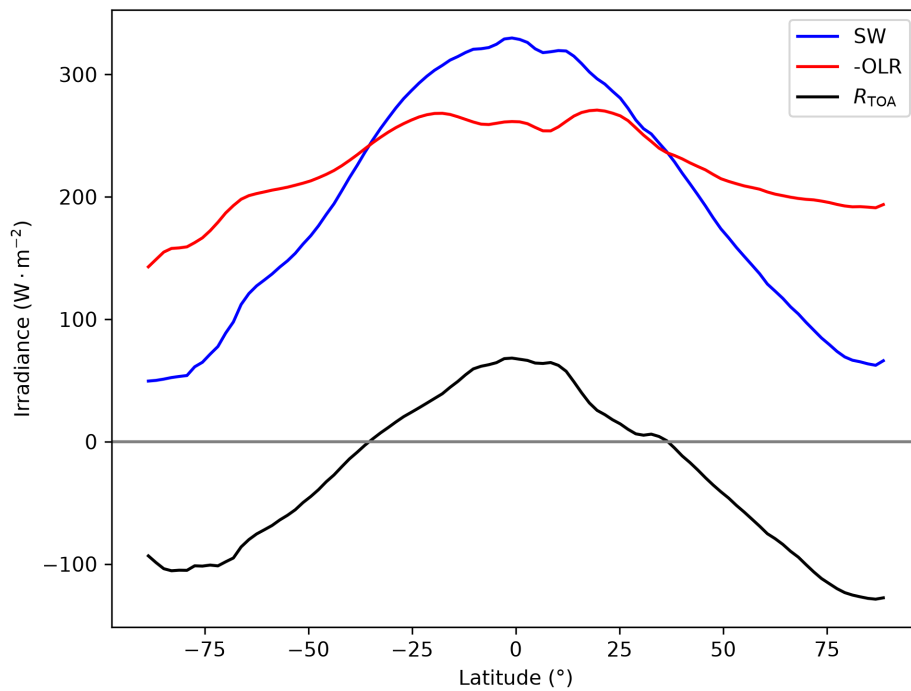


Figure 2.3

For every column of the earth system, the energy budget can be written as (Hartmann, 2016; Oort & Haar, 1976)

$$\frac{\partial E_{ao}}{\partial t} = R_{TOA} - \Delta F_{ao} \quad (2.13)$$

where R_{TOA} is the net radiative flux at the top of the atmosphere, which consists of the net shortwave radiation plus the net longwave radiation (positive sign means down), and ΔF_{ao} is the divergence of energy transport by the atmosphere and ocean. Integrating this equation in the steady state over a latitude band gives

$$F_{ao}(\phi) = \int_{-\pi/2}^{\phi} \int_0^{2\pi} R'_{TOA} a^2 \cos \phi d\lambda d\phi \quad (2.14)$$

which provides a measure for the energy transport at every latitude. You want to use the anomalous net radiative TOA flux R'_{TOA} here, where you have the global mean value subtracted, to make sure that the energy transport is closed (zero transport at both poles) (Zelinka & Hartmann, 2012). If you want to separate the energy transport by the atmosphere and the ocean, you can subtract the surface energy fluxes, which can be obtained from the sum of surface latent, sensible heat, and net radiative flux, from the TOA radiative fluxes.

To determine the energy transport by the Hadley cell we need the energy within an air parcel in the atmosphere. This is given by the moist static energy (MSE) which is defined as

$$h = c_p T + gz + L_v q \quad (2.15)$$

with c_p being the specific heat capacity at constant pressure, g the gravitational acceleration, z the height, L_v the latent heat of vaporization and q the specific humidity. The transport of

MSE by the mean meridional circulation at a certain latitude ϕ is given by vertical integration of the meridional wind $\bar{v}$ multiplied by the MSE h over the entire atmospheric column

$$F_{\text{HC}}(\phi) = \frac{2\pi a \cos \phi}{g} \int_0^{p_s} [\bar{v}_{\text{adj}}][\bar{h}] dp. \quad (2.16)$$

This expression compares the poleward energy transport by the upper branch of the HC to the equatorward energy transport by the lower branch. We use the adjusted meridional wind $\bar{v}_{\text{adj}}$ (really necessary?) Expressing the HC strength ψ_{max} in terms of the energy transport F_{HC} and the gross moist stability H gives

$$F_{\text{HC}} = \psi_{\text{max}} H \quad (2.17)$$

Gross moist stability (GMS) is defined as the ratio of energy transport to mass transport by the HC which can be understood as the energy transport efficiency. It is inferred to

$$H(\phi) = \frac{2\pi a \cos \phi}{g} \frac{1}{\psi_{\text{max}}} \int_0^{p_s} [\bar{v}][\bar{h}] dp. \quad (2.18)$$

It is a measure for the stratification of the atmosphere. A large GMS indicates a strongly stratified atmosphere. Trying to understand the changes in HC strength, we can linearize Equation (2.17) and obtain

$$\frac{\delta \psi_{\text{max}}}{\psi_{\text{max}}} = \frac{\delta F_{\text{HC}}}{F_{\text{HC}}} - \frac{\delta H}{H}. \quad (2.19)$$

This gives us the contribution of GMS and HC transport changes. Knowing the total energy transport in the atmosphere F_a from Equation (2.14), it can be further split up into a mean F_{HC} and an eddy energy flux component F_e , which gives

$$F_a = F_{\text{HC}} + F_e. \quad (2.20)$$

The eddy component can be calculated as the residual or by using the transient eddy fluxes of MSE

$$F_e(\phi) = \frac{2\pi a \cos \phi}{g} \int_0^{p_s} [\overline{v'h'}] dp. \quad (2.21)$$

Inserting this into the linearized Equation (2.17) gives

$$\frac{\delta \psi_{\text{max}}}{\psi_{\text{max}}} = \frac{\delta F_a}{F_{\text{HC}}} - \frac{\delta F_e}{F_{\text{HC}}} - \frac{\delta H}{H}.$$

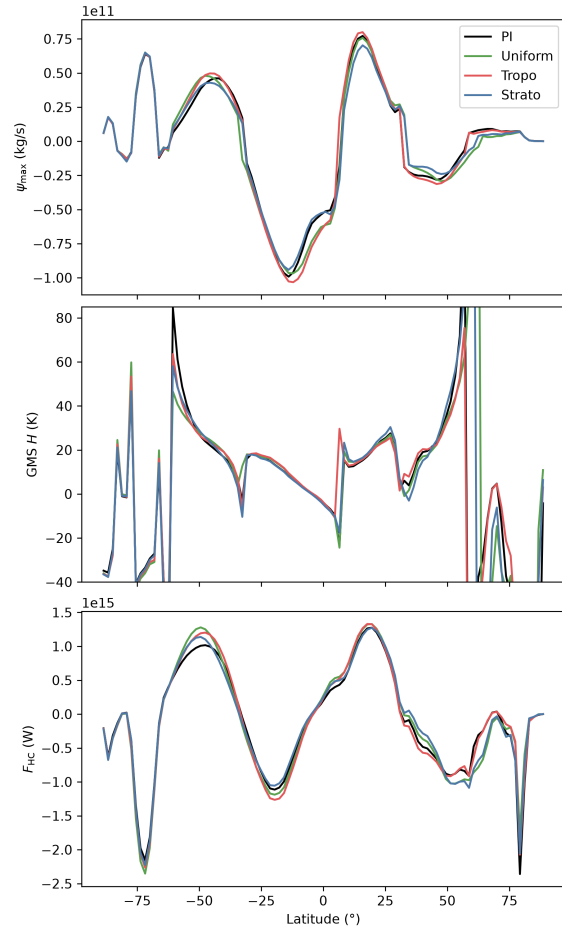


Figure 2.4: TODO: Only include PI values, increase axis labels

The HC changes are now a function of the total change in energy transport. This can be further split up into a change in radiative forcing R_f and a change due to feedbacks $\sum_i \lambda_i \Delta T_s$, which gives

$$\Delta R_{\text{TOA}} = \sum_i \lambda_i \Delta T_s + R_f.$$

The forcing is the direct effect of CO₂ on the TOA radiative fluxes and the feedbacks are the changes in TOA radiative fluxes due to changes in surface temperature. This change in surface temperature has an effect on the surface albedo, water vapor, clouds, and lapse rate, and OLR. An effect that increases surface temperature is called a positive feedback and an effect that decreases surface temperature is called a negative feedback. Positive feedbacks are the water vapor, surface albedo, and the longwave cloud radiative effect. Negative feedbacks are the Planck feedback, lapse rate feedback, and the shortwave cloud radiative effect.

$$\frac{\delta \psi_{\text{max}}}{\psi_{\text{max}}} = \frac{\delta F_{\text{forc}}}{F_{\text{HC}}} + \frac{\delta F_{\text{fdb}}}{F_{\text{HC}}} - \frac{\delta F_e}{F_{\text{HC}}} - \frac{\delta H}{H} \quad (2.22)$$

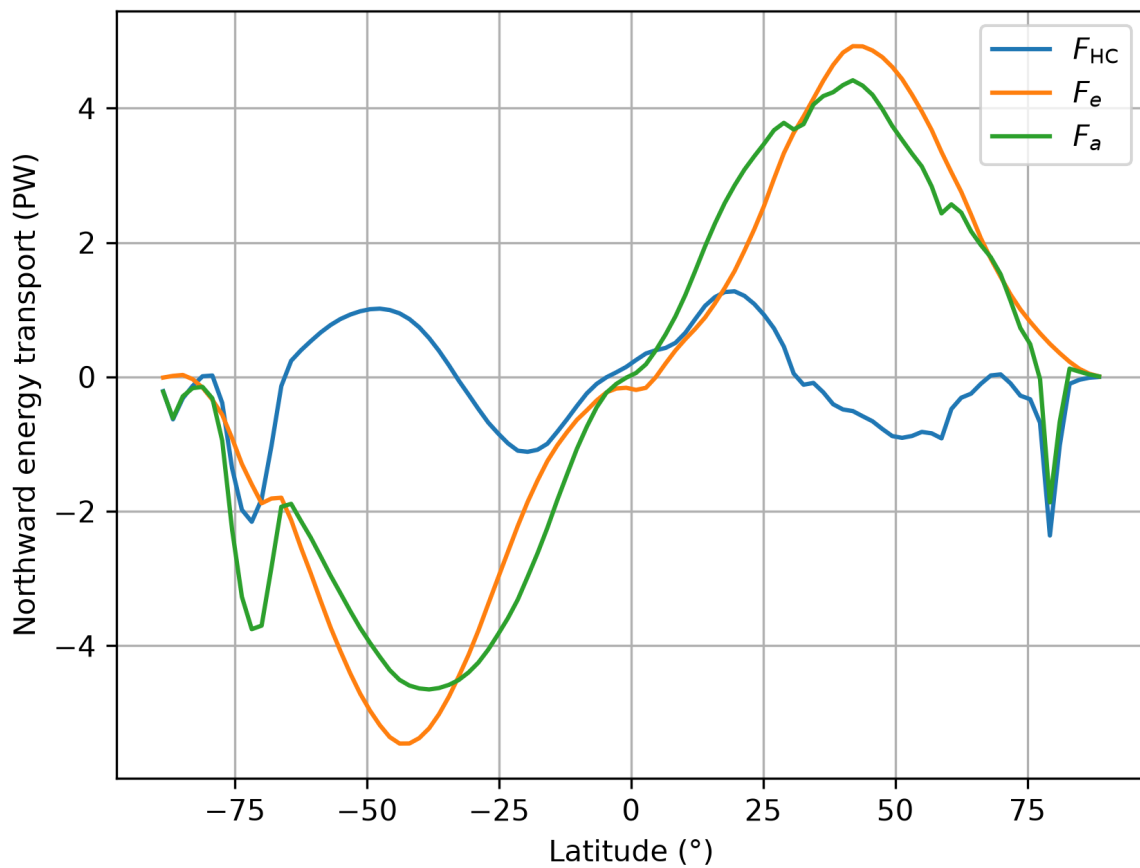


Figure 2.5

2.5. Model setup

2.5.1. ECHAM5

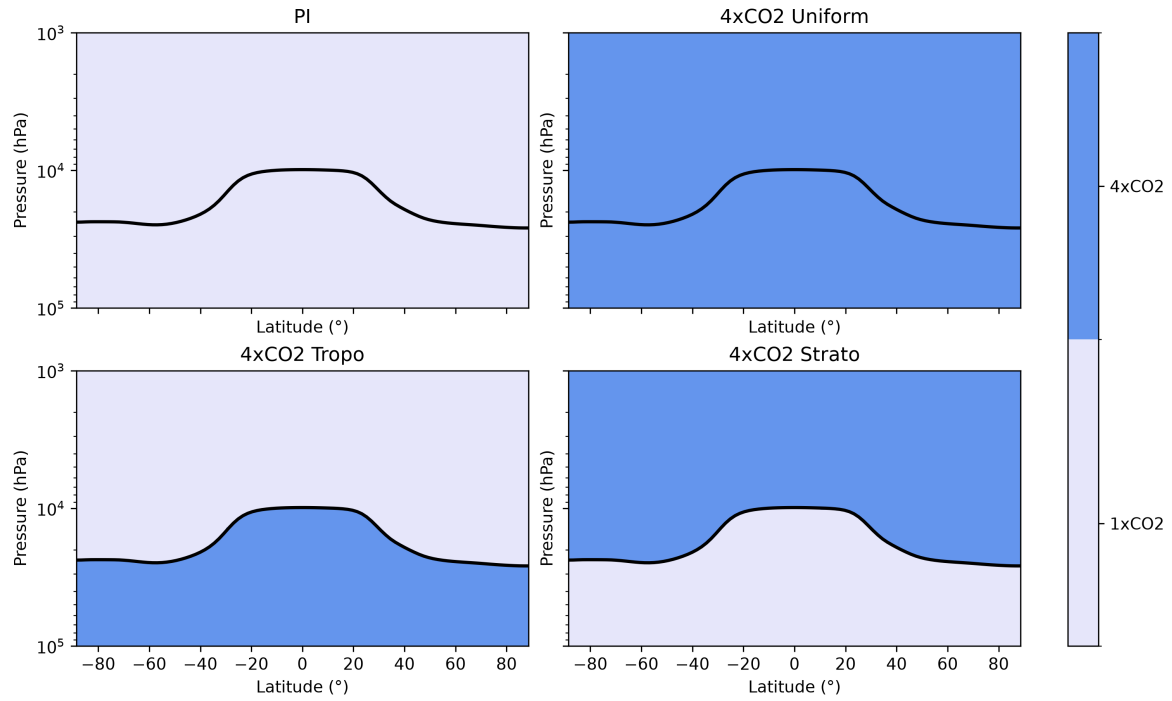


Figure 2.6

2.5.2. konrad

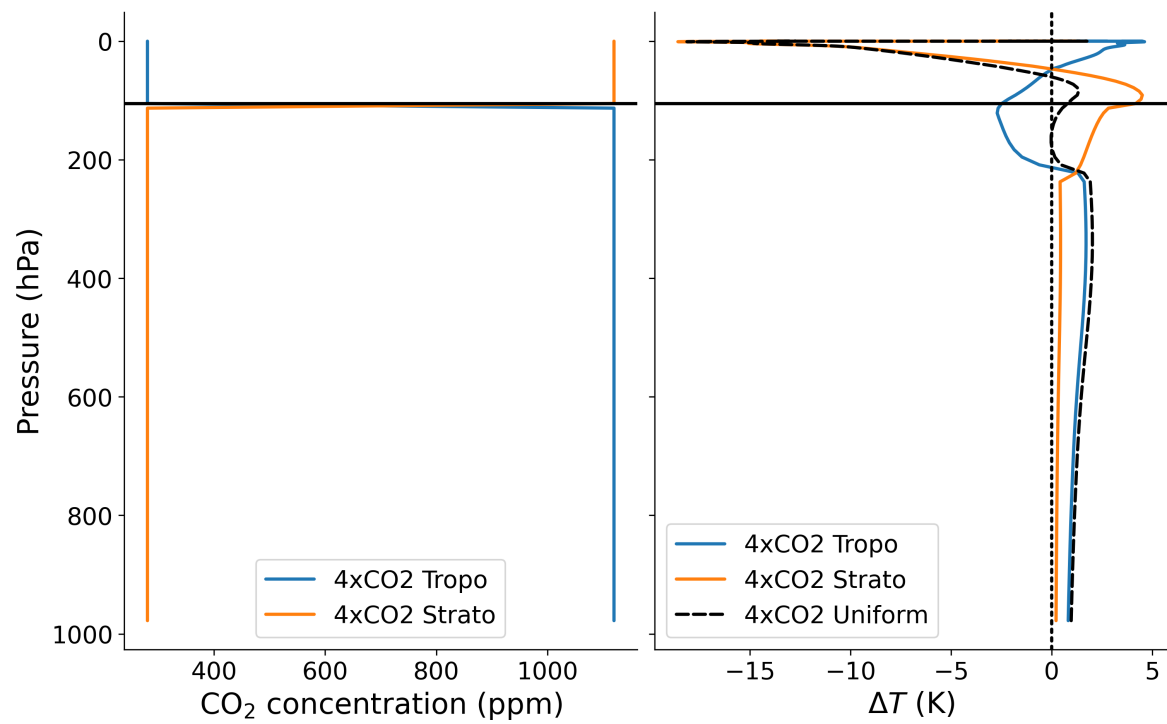


Figure 2.7

3

Methods

- 3.1. ECHAM5
- 3.2. konrad
- 3.3. Decomposition in forcing factors

4

Results

4.1. The atmospheric response

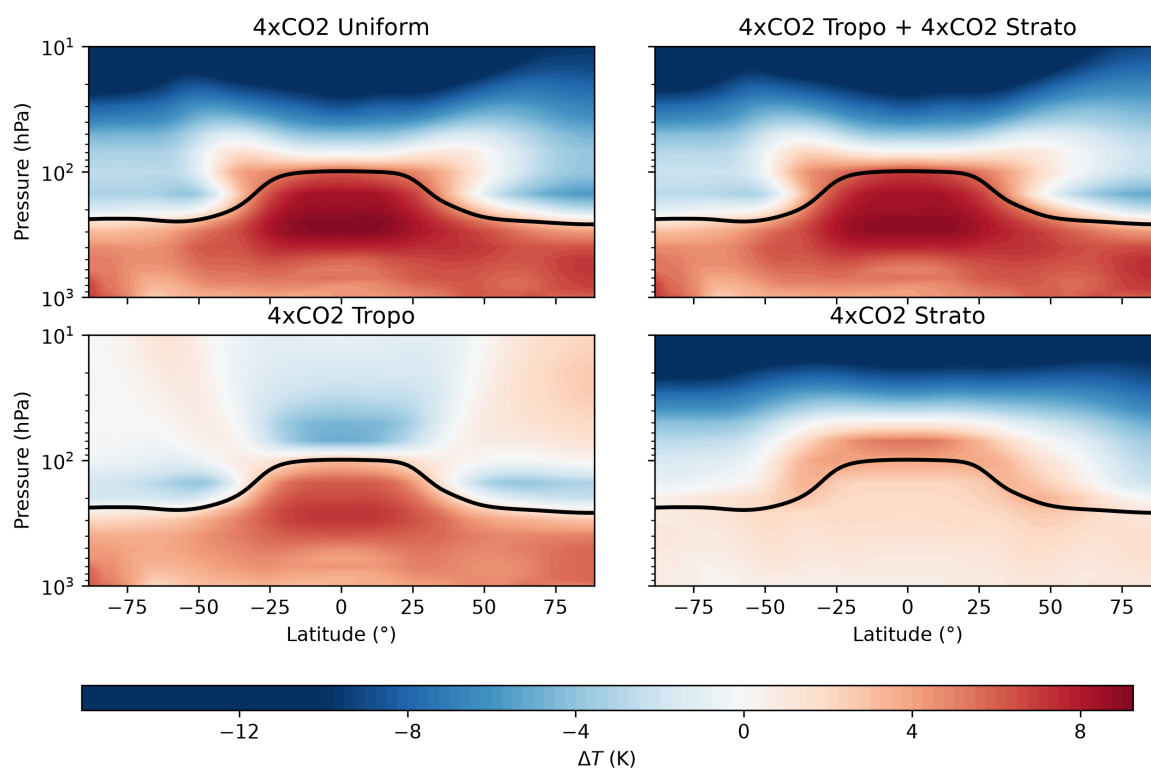


Figure 4.1

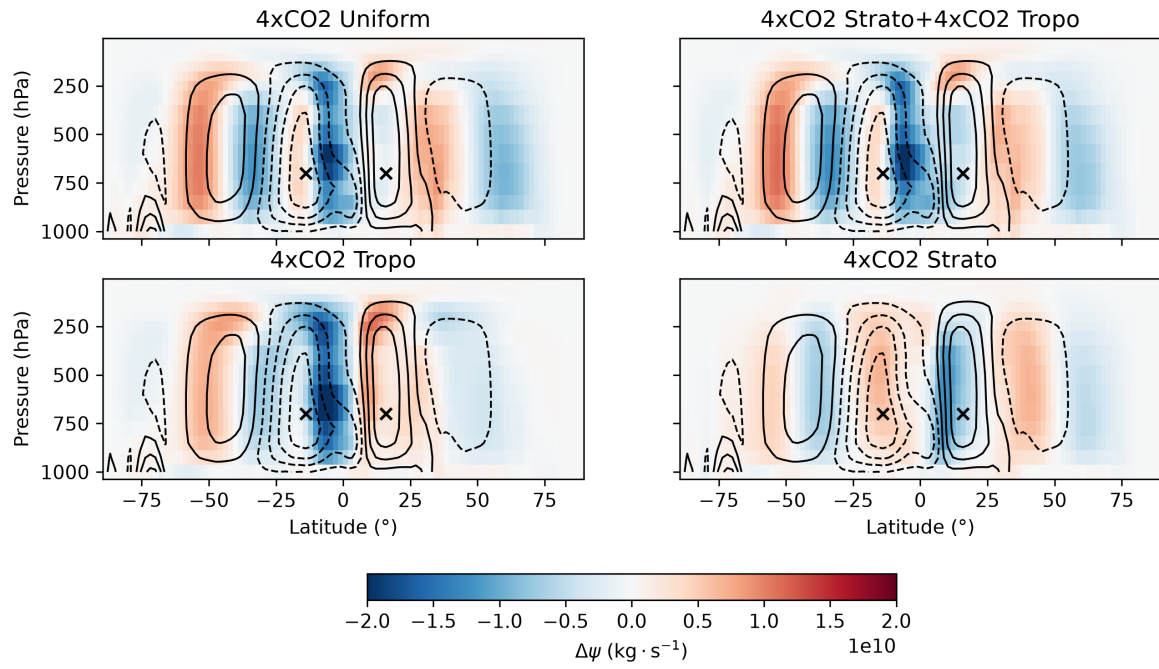


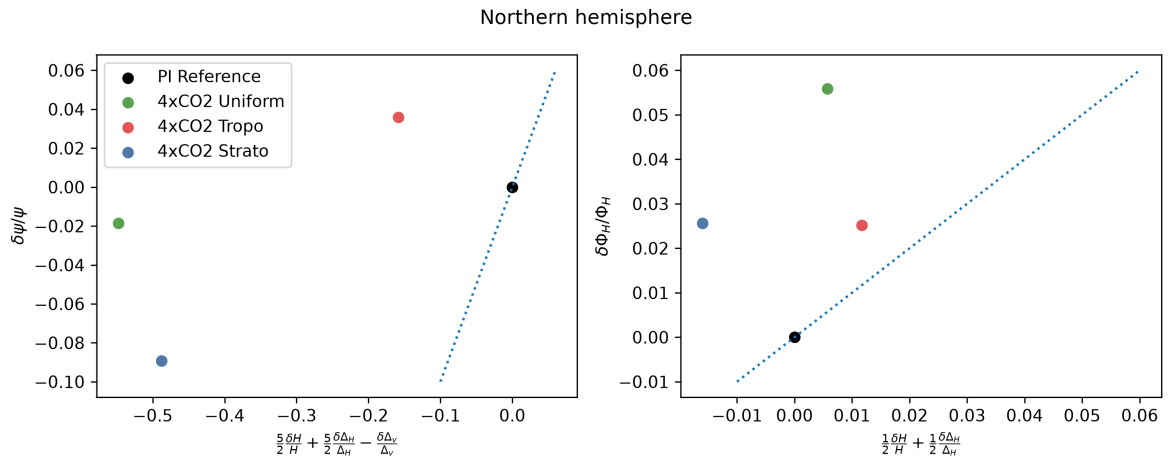
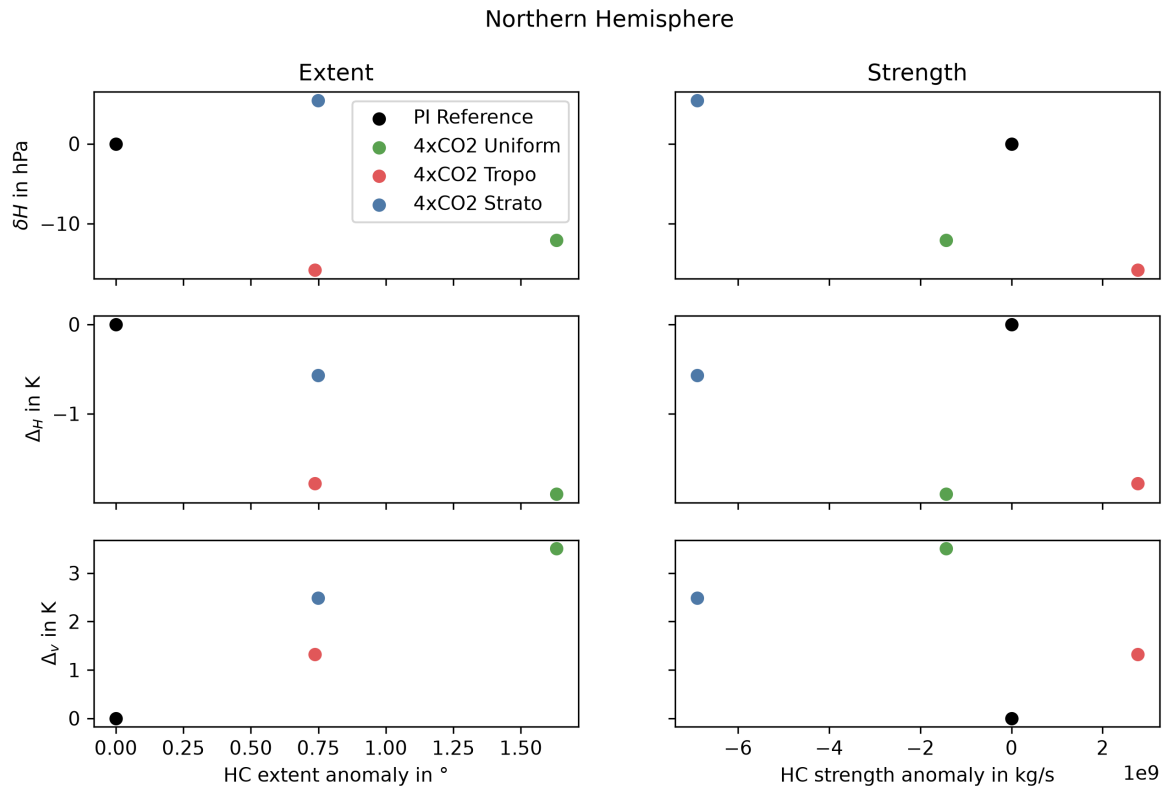
Figure 4.2

4.2. Predictions from axisymmetric scalings

(Seo et al., 2014)

$$\frac{\delta\psi_{\max}}{\psi_{\max}} = \frac{5}{2} \frac{\delta H}{H} + \frac{5}{2} \frac{\delta\Delta_h}{\Delta_h} - \frac{\delta\Delta_v}{\Delta_v} \quad (4.1)$$

$$\frac{\delta\phi_H}{\phi_H} = \frac{1}{2} \frac{\delta H}{H} + \frac{1}{2} \frac{\delta\Delta_h}{\Delta_h} \quad (4.2)$$



4.3. Eddy driven Hadley cell predictions

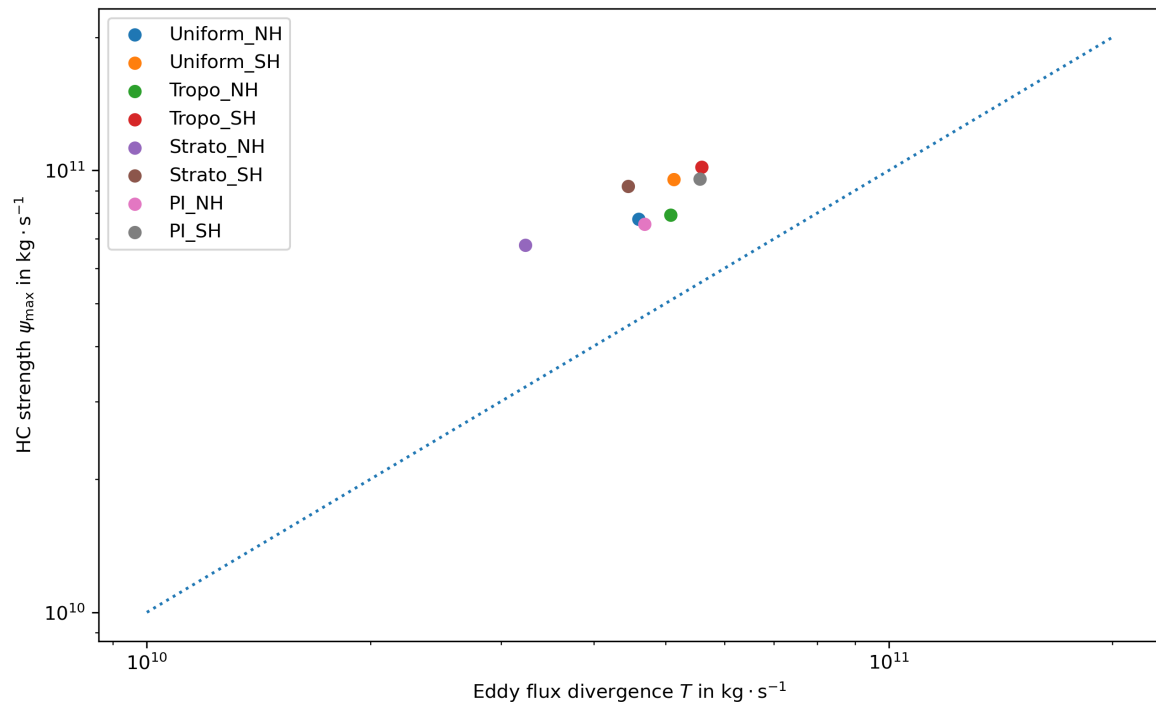


Figure 4.5

4.4. The energetic framework

4.4.1. GMS and energy transport of the Hadley cell

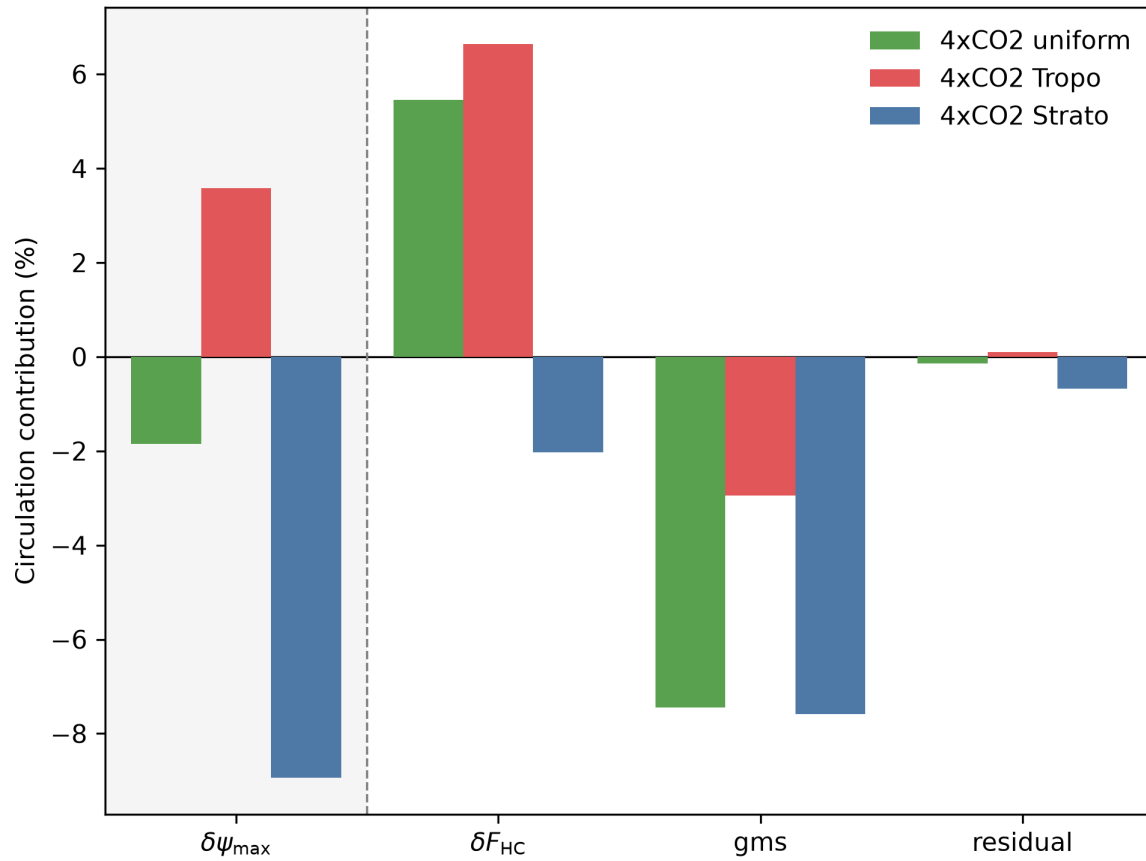


Figure 4.6

4.4.2. Eddy energy transport

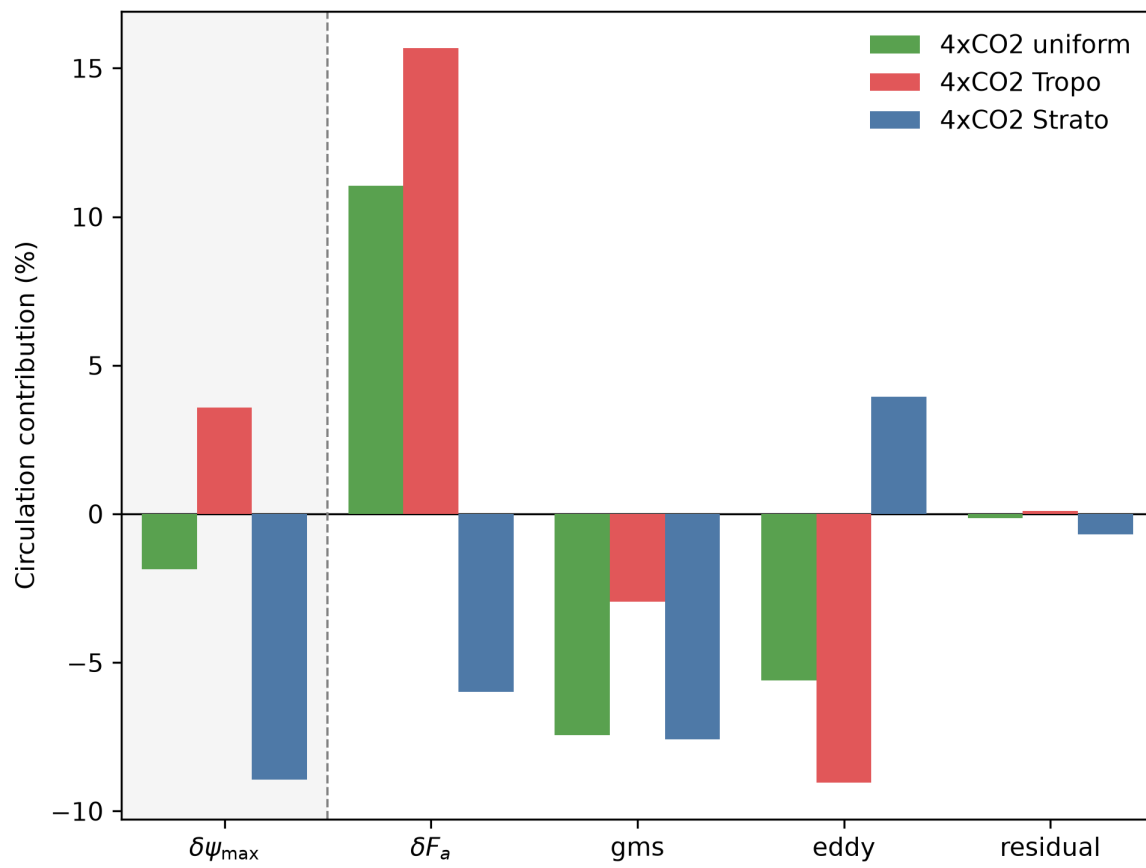


Figure 4.7

4.4.3. Forcing and feedback contributions

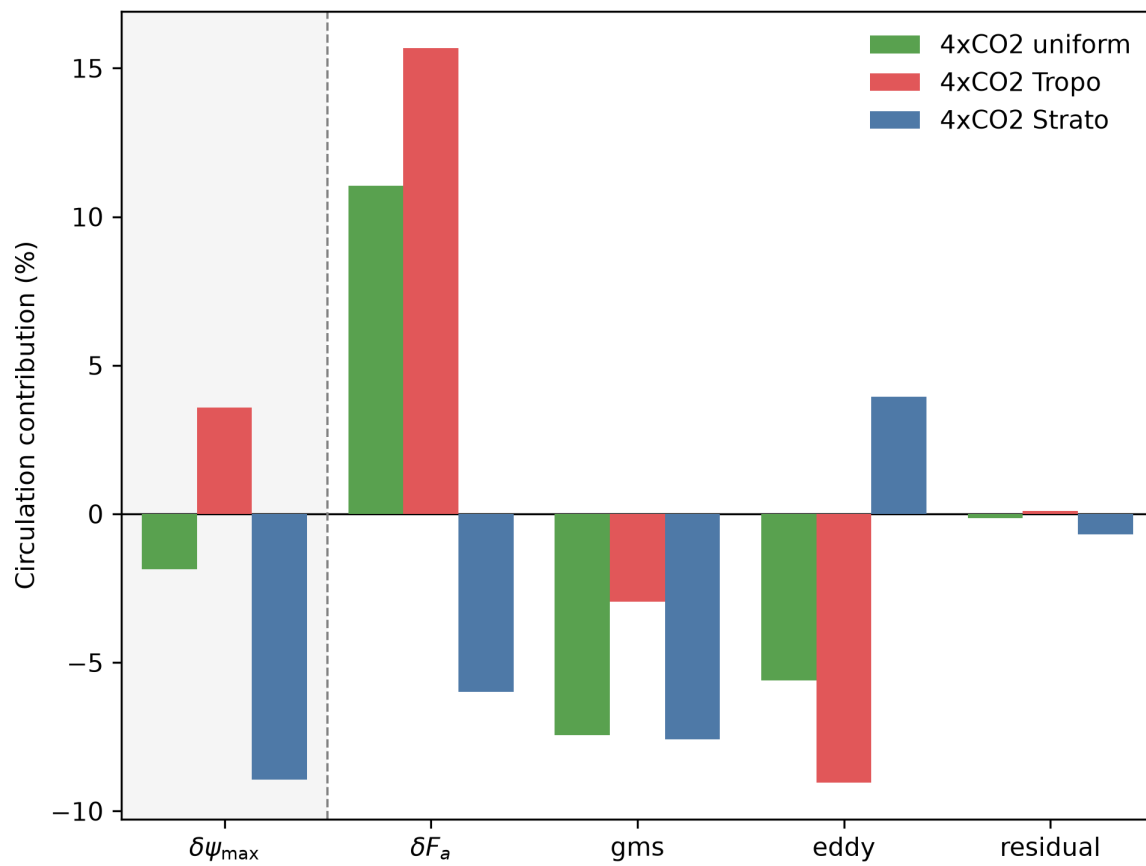


Figure 4.8

4.4.4. The forcing

Explain the forcing structure (Merlis, 2015)

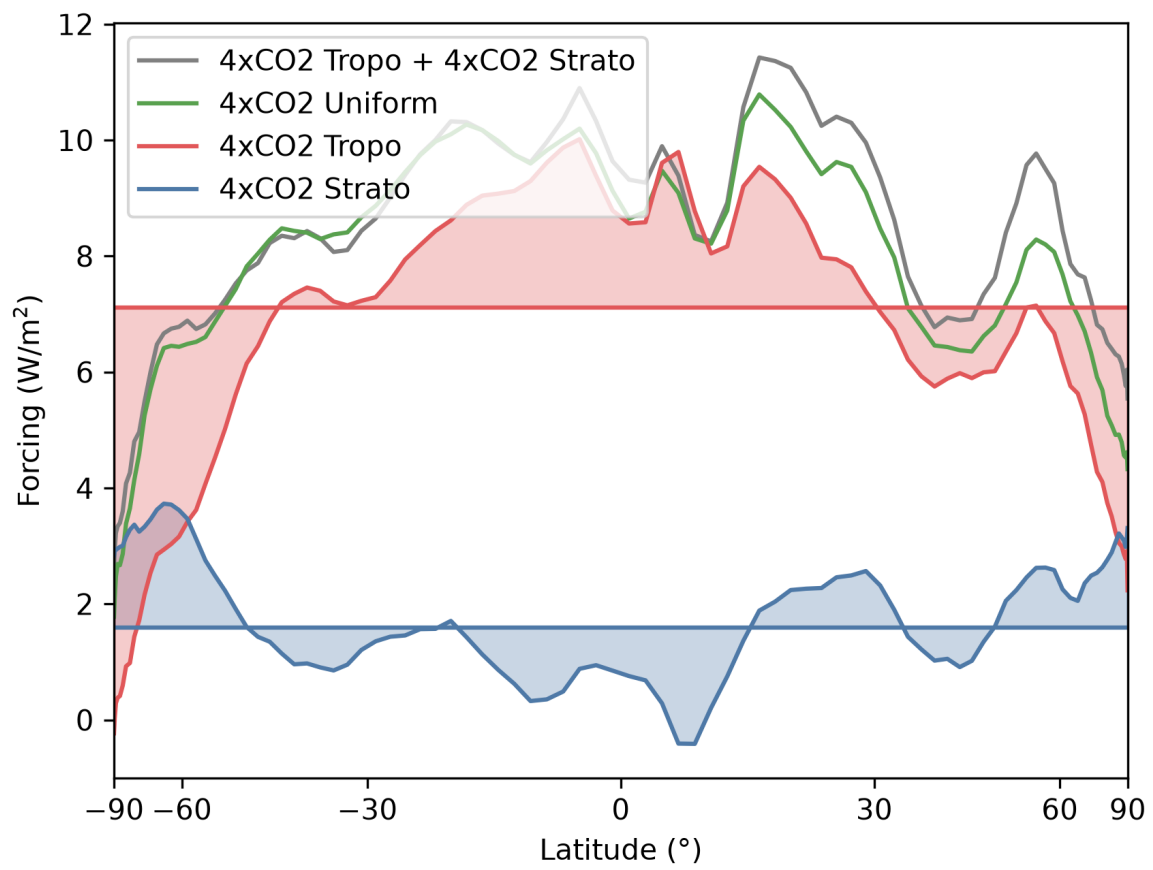


Figure 4.9

5

Discussion

6

Conclusion

Acknowledgement

First, I like to thank my first supervisor , for Also, I want to thank my second supervisor for

Thanks to for all the insights into Thanks to for all the intense but fruitful scientific debate about Thanks to ... , who always had an open door for

Also, I would like to thank the

And last but not least, thanks to , who helped As well as for the help with

References

- Bony, S., Bellon, G., Klocke, D., Sherwood, S., Fermepin, S., & Denvil, S. (2013). Robust direct effect of carbon dioxide on tropical circulation and regional precipitation. *Nature Geoscience*, 6(6), 447–451. <https://doi.org/10.1038/ngeo1799>
- Bordoni, S., & Schneider, T. (2008). Monsoons as eddy-mediated regime transitions of the tropical overturning circulation. *Nature Geoscience*, 1(8), 515–519. <https://doi.org/10.1038/ngeo248>
- Chemke, R., & Polvani, L. M. (2021). Elucidating the Mechanisms Responsible for Hadley Cell Weakening Under $4 \times \text{CO}_2$ Forcing. *Geophysical Research Letters*, 48(3), e2020GL090348. <https://doi.org/10.1029/2020GL090348>
- D’Agostino, R., Lionello, P., Adam, O., & Schneider, T. (2017). Factors controlling Hadley circulation changes from the Last Glacial Maximum to the end of the 21st century. *Geophysical Research Letters*, 44(16), 8585–8591. <https://doi.org/10.1002/2017GL074533>
- Fan, C.-S., & Dommenges, D. (2024). The weakening of the tropical circulation is caused by the lifting of the tropopause height. *Climate Dynamics*, 62(1), 187–201. <https://doi.org/10.1007/s00382-023-06909-1>
- Feldl, N., & Bordoni, S. (2016). Characterizing the Hadley Circulation Response through Regional Climate Feedbacks. *Journal of Climate*, 29(2), 613–622. <https://doi.org/10.1175/JCLI-D-15-0424.1>
- Hartmann, D. L. (2016). *Global Physical Climatology*. Elsevier. <https://doi.org/10.1016/C2009-0-00030-0>
- Held, I. M. (2000). *The general circulation of the atmosphere*. Retrieved July 9, 2026, from https://www.gfdl.noaa.gov/wp-content/uploads/files/user_files/ih/lectures/woods_hole.pdf
- Held, I. M., & Hou, A. Y. (1980). Nonlinear Axially Symmetric Circulations in a Nearly Inviscid Atmosphere. *Journal of the Atmospheric Sciences*, 37(3), 515–533. [https://doi.org/10.1175/1520-0469\(1980\)037<0515:NASCIA>2.0.CO;2](https://doi.org/10.1175/1520-0469(1980)037<0515:NASCIA>2.0.CO;2)
- Hide, R. (1969). Dynamics of the Atmospheres of the Major Planets with an Appendix on the Viscous Boundary Layer at the Rigid Bounding Surface of an Electrically-Conducting Rotating Fluid in the Presence of a Magnetic Field. *Journal of the Atmospheric Sciences*, 26(5), 841–853. [https://doi.org/10.1175/1520-0469\(1969\)026<0841:DOTAOT>2.0.CO;2](https://doi.org/10.1175/1520-0469(1969)026<0841:DOTAOT>2.0.CO;2)
- Jeevanjee, N. (2023). Climate sensitivity from radiative-convective equilibrium: A chalkboard approach. *American Journal of Physics*, 91(9), 731–745. <https://doi.org/10.1119/5.0135727>

- Kim, D., Kim, H., Kang, S. M., Stuecker, M. F., & Merlis, T. M. (2022). Weak Hadley cell intensity changes due to compensating effects of tropical and extratropical radiative forcing. *npj Climate and Atmospheric Science*, 5(1), 61. <https://doi.org/10.1038/s41612-022-00287-x>
- Kuo, H.-L. (1956). Forced and free meridional circulations in the atmosphere. *Journal of the Atmospheric Sciences*, 13(6), 561–568. [https://doi.org/10.1175/1520-0469\(1956\)013<0561:FAFMCI>2.0.CO;2](https://doi.org/10.1175/1520-0469(1956)013<0561:FAFMCI>2.0.CO;2)
- Lindzen, R. S., & Hou, A. V. (1988). Hadley Circulations for Zonally Averaged Heating Centered off the Equator. *Journal of the Atmospheric Sciences*, 45(17), 2416–2427. [https://doi.org/10.1175/1520-0469\(1988\)045<2416:HCFZAH>2.0.CO;2](https://doi.org/10.1175/1520-0469(1988)045<2416:HCFZAH>2.0.CO;2)
- Lu, J., Vecchi, G. A., & Reichler, T. (2007). Expansion of the Hadley cell under global warming. *Geophysical Research Letters*, 34(6), 2006GL028443. <https://doi.org/10.1029/2006GL028443>
- Merlis, T. M. (2015). Direct weakening of tropical circulations from masked CO₂ radiative forcing. *Proceedings of the National Academy of Sciences*, 112(43), 13167–13171. <https://doi.org/10.1073/pnas.1508268112>
- Oort, A. H., & Haar, T. H. V. (1976). On the Observed Annual Cycle in the Ocean-Atmosphere Heat Balance Over the Northern Hemisphere. *Journal of Physical Oceanography*, 6(6), 781–800. [https://doi.org/10.1175/1520-0485\(1976\)006<0781:OTOACI>2.0.CO;2](https://doi.org/10.1175/1520-0485(1976)006<0781:OTOACI>2.0.CO;2)
- Pikovnik, M., Zaplotnik, Ž., Boljka, L., & Žagar, N. (2022). Metrics of the Hadley circulation strength and associated circulation trends. *Weather and Climate Dynamics*, 3(2), 625–644. <https://doi.org/10.5194/wcd-3-625-2022>
- Polvani, L. M., Waugh, D. W., Correa, G. J. P., & Son, S.-W. (2011). Stratospheric Ozone Depletion: The Main Driver of Twentieth-Century Atmospheric Circulation Changes in the Southern Hemisphere. *Journal of Climate*, 24(3), 795–812. <https://doi.org/10.1175/2010JCLI3772.1>
- Seo, K.-H., Frierson, D. M. W., & Son, J.-H. (2014). A mechanism for future changes in Hadley circulation strength in CMIP5 climate change simulations. *Geophysical Research Letters*, 41(14), 5251–5258. <https://doi.org/10.1002/2014GL060868>
- Singh, M. S., Kuang, Z., & Tian, Y. (2017). Eddy Influences on the Strength of the Hadley Circulation: Dynamic and Thermodynamic Perspectives. *Journal of the Atmospheric Sciences*, 74(2), 467–486. <https://doi.org/10.1175/JAS-D-16-0238.1>
- Vallis, G. K. (2017). *Atmospheric and Oceanic Fluid Dynamics: Fundamentals and Large-Scale Circulation* (2nd ed.). Cambridge University Press. <https://doi.org/10.1017/9781107588417>
- Vallis, G. K., Zurita-Gotor, P., Cairns, C., & Kidston, J. (2015). Response of the large-scale structure of the atmosphere to global warming. *Quarterly Journal of the Royal Meteorological Society*, 141(690), 1479–1501. <https://doi.org/10.1002/qj.2456>
- Walker, C. C., & Schneider, T. (2005). Response of idealized Hadley circulations to seasonally varying heating. *Geophysical Research Letters*, 32(6), 2004GL022304. <https://doi.org/10.1029/2004GL022304>

- Walker, C. C., & Schneider, T. (2006). Eddy Influences on Hadley Circulations: Simulations with an Idealized GCM. *Journal of the Atmospheric Sciences*, 63(12), 3333–3350. <https://doi.org/10.1175/JAS3821.1>
- Xia, Y., Hu, Y., & Liu, J. (2020). Comparison of trends in the Hadley circulation between CMIP6 and CMIP5. *Science Bulletin*, 65(19), 1667–1674. <https://doi.org/10.1016/j.scib.2020.06.011>
- Zelinka, M. D., & Hartmann, D. L. (2012). Climate Feedbacks and Their Implications for Poleward Energy Flux Changes in a Warming Climate. *Journal of Climate*, 25(2), 608–624. <https://doi.org/10.1175/JCLI-D-11-00096.1>

Eidesstattliche Versicherung

Hiermit versichere ich an Eides statt, dass ich die vorliegende Arbeit im Studiengang M.Sc. Atmospheric Science selbstständig verfasst und keine anderen als die angegebenen Hilfsmittel – insbesondere keine im Quellenverzeichnis nicht benannten Internet-Quellen – benutzt habe. Alle Stellen, die wörtlich oder sinngemäß aus Veröffentlichungen entnommen wurden, sind als solche kenntlich gemacht. Ich versichere weiterhin, dass ich die Arbeit vorher nicht in einem anderen Prüfungsverfahren eingereicht habe. Sofern im Zuge der Erstellung der vorliegenden Abschlussarbeit auf generativer künstlicher Intelligenz (gKI) basierte elektronische Hilfsmittel verwendet wurden, versichere ich, dass meine eigene Leistung im Vordergrund stand und dass eine vollständige Dokumentation aller verwendeten Hilfsmittel gemäß der Guten Wissenschaftlichen Praxis vorliegt. Ich trage die Verantwortung für eventuell durch die gKI generierte fehlerhafte oder verzerrte Inhalte, fehlerhafte Referenzen, Verstöße gegen das Datenschutz- und Urheberrecht oder Plagiate. Einer Veröffentlichung der vorliegenden Arbeit in der zuständigen Fachbibliothek des Fachbereichs stimme ich zu.

Hamburg, den _____ Unterschrift: _____